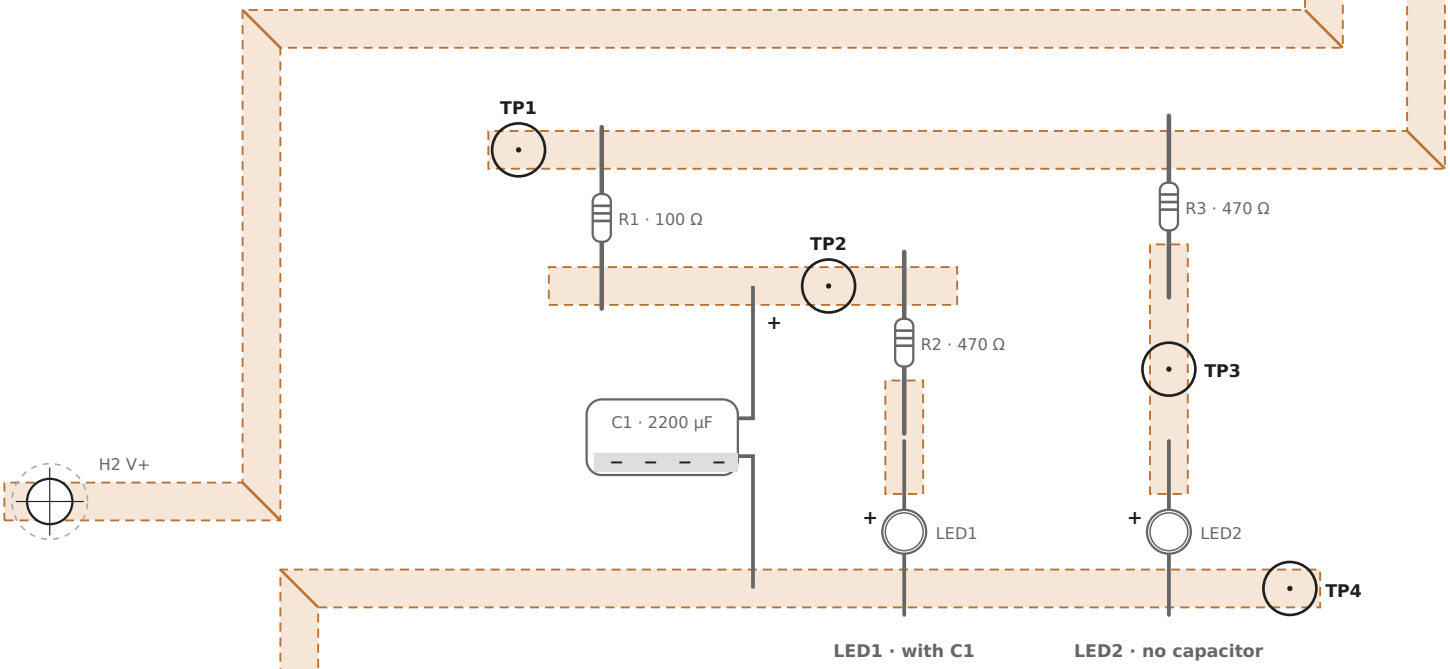
L08 Store some charge

A capacitor keeps an LED lit after the power goes. For how long?

Draft: tagged steps may change

H1 BUS A · NC

S1
fold the corner down



H2 V+

H3 GND

You need

- 2 red LEDs, 5 mm
- R1: 100 Ω (brown-black-brown) **[P]**
- R2, R3: 470 Ω (yellow-violet-brown) **[P]**
- C1: 2200 µF electrolytic capacitor, 10 V or more
- 5 mm copper tape; P1 in the binder; a multimeter

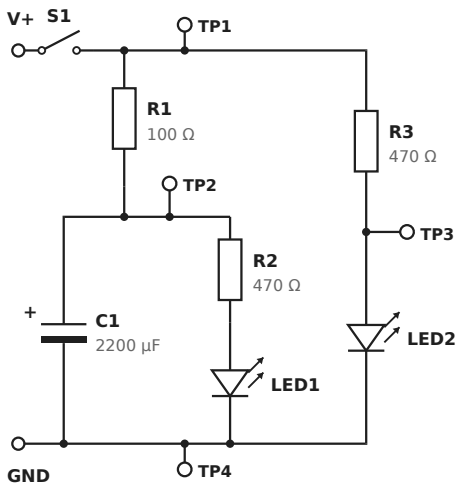
Build it

1. Lay the six lanes, one piece of tape each. Fold at the creases; never cut.
2. Stick a 10 mm piece of copper tape on the corner flap, S1. **[T0]**
3. Punch the holes and set eyelets in H2 and H3. **[T0]**
4. Place R1, R2 and R3 across their gaps, either way round.
5. Place C1 **exactly as printed**: the striped leg (−) down to GND, the other leg up to the lane. Backwards, a capacitor can heat up and fail.
6. Place LED1 and LED2 as printed: long leg on the +.
7. Hold each lead flat on its lane with a small patch of copper tape. **[T0]**
8. Bind with P1. Hold the corner down, then let go and watch both LEDs. If C1 ever feels warm, switch BT1 off and check it.

[T0] and **[P]** mark choices that may change: **[T0]** after the contact tests, **[P]** once the power supply is settled.

H4 BUS B · NC

Schematic



C1 charges while S1 is held down.

Predict When you let go of the corner, what will each LED do?

LED1 goes out at once / fades **LED2** goes out at once / fades

Circle one for each. Why do you think so?

Measure Count the seconds each LED glows after you let go.

Press	LED1	LED2	Black probe on TP4:
A quick tap	____ s	____ s	TP2 when LED1
Hold 1 s	____ s	____ s	goes dark ____ V
Hold 5 s	____ s	____ s	

Explain Where does LED1's light come from after you let go?

How could you make it glow for longer?

Print at 100 %: this bar should measure 100 mm.